⌈Quotient Space

⌈Def. Let $(X, \mathcal{T})$ be a Topological space, and Y be a set.

For an onto map $f: X \to Y$, define a Quotient Topology on Y induced by f as

$$\mathcal{T}_Q \overset{\text{def}}{=} \{ U \subseteq Y \mid f^{-1}[U] \text{ is open of } (X, \mathcal{T}) \},$$

It is actually a Topology on Y, because: clearly, $\emptyset, Y \in \mathcal{T}_Q$ and

$U_\alpha \in \mathcal{T}_Q \ (\alpha \in I) \Rightarrow f^{-1}\left[\bigcup_{\alpha \in I} U_\alpha\right] = \bigcup_{\alpha \in I} f^{-1}[U_\alpha]$ is also open in X, so is $\bigcup U_\alpha$.

Similarly, $f^{-1}\left[\bigcap_{\alpha \in I} U_\alpha\right] = \bigcap_{\alpha \in I} f^{-1}[U_\alpha]$ also open if I is finite, so is $\bigcap U_\alpha$.⌋

Moreover, it is the finest Topology on Y such that f is Continuous, because

$f: X \to Y$ is Continuous iff $\forall U \in \mathcal{T}_Y, f^{-1}[U] \in \mathcal{T}_X$ iff $\mathcal{T}_Y \subseteq \mathcal{T}_Q$.

where $\mathcal{T}_Y$ is arbitrary Topology on Y. ☒

⌈Def. Let X be a Topological Space, and $\sim$ be an equivalence relation on X.

Define the Canonical map on X as

$$\pi : X \to X/\sim ; x \mapsto [x]$$

The $(X/\sim, \mathcal{T}_\pi)$ is called Quotient Space where $\mathcal{T}_\pi$ is a Quotient topology on $X/\sim$, induced by π.

Let X be a Topological Space, $\sim$ be an equivalent relation on X, π be a Canonical map.

「Lemma. Let Z be a Topological Space and a map $g: X/\sim \to Z$,

g is Continuous if and only if $\underset{: X \to Z}{g \circ \pi}$ is Continuous.

Proof. Since $\pi : X \to X/\sim$ is Continuous, trivially $g \circ \pi$ is Continuous.

Conversely, assume $g \circ \pi$ is Continuous. Then, given an open $U \subseteq Z$,

$(g \circ \pi)^{-1}[U] = \pi^{-1}[g^{-1}[U]]$ is open of X, so $g^{-1}[U]$ is open of $X/\sim$. ∎

「Lemma. Let Z be a Topological Space.

If a Continuous map $f : X \to Z$ satisfies $x \sim y \Rightarrow f(x) = f(y)$,

then a map induced by f:

$$\bar{f} : X/\sim \to Z : [x] \mapsto f(x)$$

is Continuous and it is a Unique map such that $\bar{f} \circ \pi = f$.

Proof. $\bar{f}$ is well-defined by: $[x] = [y] \Leftrightarrow x \sim y \Rightarrow f(x) = f(y)$.

Given $x \in X$, $f(x) = \bar{f}([x]) = \bar{f}(\pi(x)) = (\bar{f} \circ \pi)(x)$, so $\bar{f} \circ \pi = f$

By the Lemma above, $\bar{f}$ is Continuous, And, given a map $g : X/\sim \to Z$ s.t. $g \circ \pi = f$,

$$g([x]) = g(\pi(x)) = f(x) = \bar{f}([x])$$

So the Uniqueness is guaranteed.

「Lemma. Let Z be a Topological Space.

If a Continuous onto map $f : X \to Z$ satisfies $x \sim y \Leftrightarrow f(x) = f(y)$ and either open or closed,

then a map induced by f:

$$\bar{f} : X/\sim \to Z : [x] \mapsto f(x)$$

is Homeomorphism.

⌈Quotient map.

⌈Def. Given Topological space X and Y, a map $f: X \to Y$ is called a quotient map if: f is Continuous, Onto, and satisfies for any $U \subseteq Y$,

$$U \text{ is open in } Y \underset{\text{Quotient}}{\overset{f \text{ conti}}{\Longleftrightarrow}} f^{-1}[U] \text{ is open in } X.$$

⌈Prop. Composition of Quotient map is also Quotient map.

Proof Let $f: X \to Y$ and $g: Y \to Z$ be Quotient maps. Given open set $U \subseteq Z$,

$$(g \circ f)^{-1}[U] \text{ is open} \Leftrightarrow f^{-1}[g^{-1}[U]] \text{ is open} \Leftrightarrow g^{-1}[U] \text{ is open} \Leftrightarrow U \text{ is open}$$

And, $(g \circ f)[X] = g[f[X]] = g[Y] = Z$, $(g \circ f)$ is a Quotient map. ∎

⌈Prop. If a Continuous Onto map $f: X \to Y$ is either open or closed map, then f is Quotient map.

Proof Given $U \subseteq Y$, if $f^{-1}[U]$ is open then $f[f^{-1}[U]] \overset{\text{onto}}{=} U$ is open, if f is open map.
And, if $f^{-1}[U]$ is open, then $f[X \setminus f^{-1}[U]]$ is closed if f is closed map.
Since f is onto, $X \setminus f^{-1}[U] = f^{-1}[Y] \setminus f^{-1}[U] = f^{-1}[Y \setminus U]$, so

$$f[X \setminus f^{-1}[U]] = f[f^{-1}[Y \setminus U]] = Y \setminus U \text{ is closed, or } U \text{ is open.}$$ ∎

But, there is a quotient map neither open nor closed: define $X = \{(x,y) \in \mathbb{R}^2 \mid x \geq 0 \text{ or } y = 0\}$.

$$f: X \to \mathbb{R};\ (x,y) \mapsto x$$

Then, it is clearly onto, and Continuous is given by: for any $a < b$ in $\mathbb{R}$,

$$f^{-1}[(a,b)] = [(a,b) \times \mathbb{R}] \cap X \quad \text{is open.}$$

To show f is quotient map, suppose that $f^{-1}[U]$ is open for some $U \subseteq \mathbb{R}$.
Given $x \in U$, $f((x,0)) = x \in U$, so $(x,0) \in f^{-1}[U]$. Since $f^{-1}[U]$ is open, for some $\varepsilon > 0$,
$(x,0) \in B_\varepsilon((x,0)) \cap X \subseteq f^{-1}[U]$. Thus, $f[B_\varepsilon((x,0))] = (x-\varepsilon, x+\varepsilon) \cap f[X] = (x-\varepsilon, x+\varepsilon) \subseteq U$.
So, U is open.
Moreover, $B_1((0,2)) \cap X$ is open of X, but $f[B_1((0,2)) \cap X] = [0,1)$, not open.
And, $F = \{(\frac{1}{n}, n) \mid n \in \mathbb{N}\}$ is closed because $F' = \emptyset$, so $F = \overline{F}$.
But, $f[F] = \{\frac{1}{n} \mid n \in \mathbb{N}\}$ is not closed in $\mathbb{R}$, since $0 \notin f[F]$ but $0 \in f[F]'$.

Comment: A projection $\pi: \mathbb{R}^2 \to \mathbb{R}$ cannot be closed map: $F = \{(\frac{1}{n}, n) \mid n \in \mathbb{N}\}$.